

Mohamed Abu Assaf

Software Engineer

Software Engineer with 3+ years of experience building Python backend systems, data pipelines, APIs, and production infrastructure. Experienced with FastAPI, PostgreSQL, automated testing, Docker, CI/CD, and data-intensive applications, with additional full-stack and cross-platform mobile development experience using React, React Native, and TypeScript.

Experience

Software Developer

Gate Institute June 2023 - Present

Urban Sensing Data Platform

- Architected and built a multi-source Python urban-sensing platform, integrating MQTT/LoRa streams and external APIs into PostgreSQL for environmental and mobility datasets.
- Designed continuous ingestion and scheduled ETL workflows with buffering, retries, completeness validation, failure recovery, and idempotent historical backfills.
- Built modular, extensively tested Python services covering source integrations, transformations, persistence, database operations, and failure scenarios.
- Containerized production workloads using Docker and Docker Compose, with Linux/systemd services supporting continuous ingestion and scheduled processing.

City Digital Twin & Backend Applications

- Designed and maintained FastAPI/SQLAlchemy backend services for data-intensive urban and geospatial applications backed by PostgreSQL.
- Improved API performance, validation, concurrency, error handling, and reliability for time-series and visualization workloads.
- Automated build and deployment workflows using GitHub Actions, self-hosted runners, secrets management, health checks, container replacement, and post-deployment validation.
- Refactored and optimized an automated data-collection workflow, reducing execution time from approximately 2–4 hours to ~20 minutes.

Full-Stack & Mobile Development

- Built and maintained a cross-platform React Native (Expo) application in TypeScript, implementing authentication and token refresh, React Query caching, network-aware behavior, multilingual support, interactive maps, analytics dashboards, and Jest testing.
- Developed React/TypeScript web functionality for data-heavy applications using interactive maps, charts, filters, dashboards, and API-driven state management.

Contact

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Skills

Backend & Data

Python · FastAPI · Flask ·
PostgreSQL · SQL · SQLAlchemy
· Pydantic · REST APIs · ETL ·
MQTT · Pandas · Selenium

DevOps & Infrastructure

Docker · Docker Compose · GitHub
Actions · CI/CD · Linux · systemd ·
Git · Azure · Terraform · Ansible ·
Bash

Frontend & Mobile

TypeScript · JavaScript · React ·
React Native · Expo · React
Query · Tailwind CSS · Jest

Education

New Bulgarian University

Bachelor of Telecommunications
and Computer Technologies

Software University (SoftUni)

Practical coursework in backend
development, APIs, databases, full-
stack development, and
DevOps/cloud technologies.

Languages

English (Advanced)
Arabic (Native or Fluent)
Bulgarian (Native or Fluent)